Compostable Garments via Digital Fabrication: a Nursing Bra Case Study

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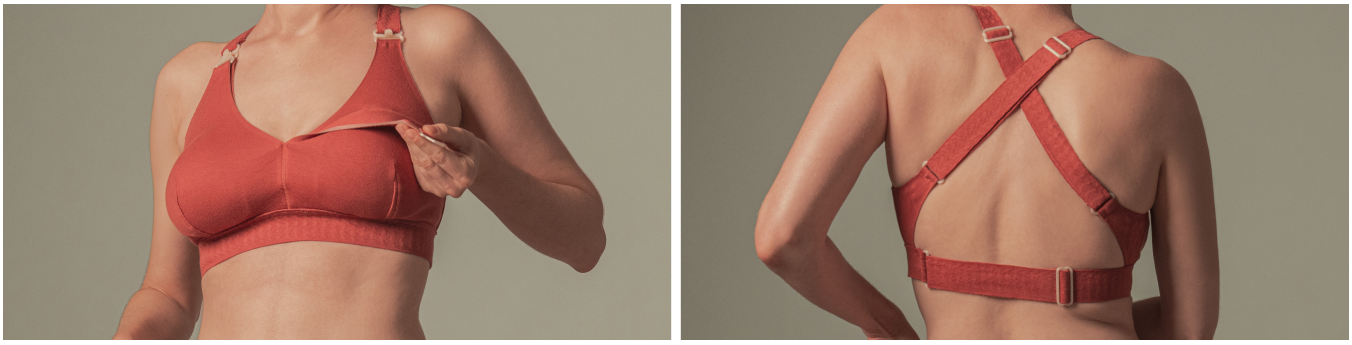


Figure 1: Nursing bra prototype made of 100 percent cotton interlock knitted fabric and biodegradable filament.

Abstract

Post-consumer textile waste is growing each year due to over-consumption and challenges in reusing, recycling, and composting garments, as blended fabrics and added components such as elastic bands, clasps and closures make material separation difficult. HCI researchers can play a central role in tackling this problem by proposing new approaches to integrate novel sustainable materials, such as biodegradable bioplastics, to create sustainable garments that are scalable for production. In this paper, we employ a material-driven design approach to develop sustainable garment components from biodegradable bioplastic filaments, leveraging parametric design tools and digital fabrication methods. Using a nursing bra as a test case for prototyping, the development of three garment components is demonstrated: an elastic band with a printed auxetic structure, a heat-bonded elastic stabilizer, and strap adjustment sliders and nursing clips. The resulting prototype is a comfortable, functional nursing bra that is fully biodegradable in an industrial composting facility.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.



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Keywords

Textile design, Digital fabrication, Parametric design, Sustainability, Material-driven design, Fashion design

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1 Introduction

The global garment industry is facing a turning point in its practices, with the need to meet global sustainability goals challenging a culture of over-consumption and the industry's dependence on non-biodegradable materials. Garment consumption is expected to reach a total of 102 million tons per year in 2030 [3], with only 1 percent of these garments being recycled, and close to 60 percent of garments are blended with synthetic fibers that are non-biodegradable [3]. The complexity of handling all this waste stems from the materials and construction of the garments themselves, such as elastic athletic wear and undergarments that contain a multitude of non-biodegradable materials to help achieve their desired qualities despite their relatively short life span. For example, nursing bras are a popular choice for breastfeeding mothers, who go through several nursing bras per nursing period, mainly due to body fluctuations. Most of these nursing bras contain clips and clasps from plastic and metals for the nursing mechanism and are commonly made of elastic fabrics. Elastane, a flexible polyurethane-based copolymer, is the most common material used for elastic fabrics, blended with both natural and synthetic fibers to improve the elastic recovery of the fabric for garment form fitting and longevity [4]. These

elastane-blended fabrics are difficult to recycle and compost, as fiber separation is nearly impossible [16]. Ultimately, these nursing bras are unsuitable for composting or recycling and end up in landfills or incineration.

To avoid directing post-consumer textile waste to landfills and incineration, reuse and recycling methods are utilized, as well as the less commonly used option of textile composting. The prevalence of blended fabrics, elastic bands, adhesives, and other functional components creates hurdles for these methods [18], and elastane in particular disrupts most recycling processes even at relatively low percentages [5]. Composting textiles can be a promising way to create enriched soil and other useful materials from textile waste, but it is currently only applicable for waste composed of bio-based and natural fibers such as viscose or cotton. The effort to manage and reduce textile waste should combine multiple waste management strategies, but the role of designers is central to changing the way garments are produced while still meeting the industry and consumers' needs.

Digital design tools have transformed the way fashion designers create garments, and the field of HCI has been steadily delving into the development of tools for designers to create garments that are more interactive [20], inclusive [23], and sustainable [10]. While some of these tools can be the basis for novel garment concepts and uses, they don't address the issues concerning the fabrication methods and materials that are used to give garments the qualities they already require. While textile, garment and material engineers constantly work for these sustainable alternatives, the field of HCI can be a catalyst for embedding these new technologies directly into the workflow of designers.

In this paper, we utilize digital fabrication and parametric design tools to create garments that have the end of their life in mind by integrating novel materials as components that can replace those currently in use. The material used throughout all developments is bioplastic, a class of man-made plastic-like materials made from renewable biomass, capable of safely biodegrading in industrial compost facilities [1]. A material-driven design methodology was used [12], in which bioplastic filaments that hold a standard for biodegradation in industrial compost and knitted 100% cotton fabric were the basis to develop functional textile components for garments, using a standard fused deposition modeling (FDM) printer, laser cut machine and parametric design tools. These components include an elastic band and strap, flexible heat-bonding lamination, and adjustment and closure clasps. As a case study for this approach, a nursing bra prototype was chosen for development. By creating a fully biodegradable nursing bra that doesn't compromise its functionality, we demonstrate a design process that can prototype new sustainable material applications in the fashion industry early on in the design stage, with a potential for scalable production.

2 Related work

2.1 Sustainability research in HCI

SHCI (sustainable human computer interaction) is now an established subfield of HCI research [6], covering a wide variety of areas and especially the field of responsible production and consumption [11]. Given the textile sector's prominence, it is no surprise that SHCI research seeks to improve its practices through design. Work can be found on the subjects of designing sustainable smart textiles and electronic wearables through disassembly [24] and incorporating biomaterials [21], methods of crafting mono-material and adaptable soft objects [14], and additive manufacturing of textiles that can avoid material waste prevalent in traditional manufacturing [7][19].

2.2 Composite printed textiles for improved mechanical qualities

The incorporation of ready made textiles with additive manufacturing has also been explored, creating multi-material systems in which 3D printing can be incorporated with existing fabric to alter its mechanical qualities. Several user interfaces and workflows have been developed for creating soft 3D objects [22], as well as embedding stiff components in fabrics for interaction and functionality [8], as well as creating self-forming structures [17].

2.3 Auxetics in wearables

Work in computational design has also delved into the creation of novel structures for soft wearables using programmable auxetic structures [2]. Auxetic structures, which are a class of geometries that have unique qualities such as negative Poisson's ratio and tuneable extensibility, have been used to create bi-stable self-forming wearables [13], as well as improve the comfort and customization of soft and wearable electronics [15] [9]. The morphology of the auxetic shapes can be finely tuned to a specific mechanical quality and applied to a variety of materials through different fabrication techniques, thereby allowing the optimization of a material for a particular function, as well as creating a single material object with different mechanics by varying its structure [13].

3 Materials

In order to develop the components necessary for the design of a biodegradable nursing bra, two different types of bioplastic filaments were sourced, one a stiff filament called Nonoilen from the company Fillamentum, and a flexible one called Biocirflex from the company Balena. The Nonoilen filament has a shore hardness of 71D, comparable to PETG, and holds the industrial composting standard EN 13432, and would be used for the nursing clips that allow for breastfeeding without bra removal and for strap adjustment sliders. Biocirflex has a Shore hardness of 80A that holds the standard ASTM D-6866 for composting in an industrial facility and would serve as a substitute for elastane or synthetic rubber in improving the elastic recovery of the 100 percent cotton knitted interlock fabric, preventing deformation in use. This filament was chosen for its elasticity and its ability to adhere to the fabric by heat bonding.

4 Methods

4.1 Printed auxetic geometries for improved elastic recovery

The elasticity of a knitted fabric is defined by its ability to extend and retract back into its original form, allowing for form-fitting during movement and avoiding deformation over prolonged use. This elastic recovery can be achieved by incorporating materials

that are elastic in their chemical structure, such as natural or synthetic rubber or elastane, in the form of yarns that are knit into the fabric. Since flexible bioplastics are not yet comparable in chemical elasticity to these materials, a different approach was taken to incorporate them into the knitted fabric, creating auxetic structures that have mechanical elasticity and heat-bonding them to the fabric. A parametric code was created in the Grasshopper plugin for Rhino, in which the re-entrant arrowhead cell size, amount, and internal hinge angle were fully controlled, as well as the printed layer amount and printing toolpath parameters, all directly written into the G-code output without the need for mesh modeling or external slicer programs (Figure 2). This method allows for precise control of the printing parameters and the geometry, preventing the tendency of highly flexible filaments to produce unwanted stringing and nozzle jamming by limiting travel lines. The flexible filament was printed on a Flashforge 3D printer in four layers of 0.2 mm each at the width of the bra straps as well as the under-bust band, in the form of an auxetic re-entrant arrowhead with a cellular size of approximately 15 mm in width and height. In addition, a 5 mm outline was printed on either side of the auxetic geometry to allow stabilization of the fibers on the strap edges for aesthetics and prevention of fraying. The resulting print was then heat-pressed at 160 °C between two layers of the cotton interlock fabric (Figure 3), with the auxetic extension direction set along the fabric's course. The course direction of a knit fabric has the highest level of extension and deformation, therefore, the auxetic structure was set to optimize elastic recovery where it is most needed. The fabric was steam pressed lightly before heat bonding to lay flat and avoid deformation or stretching.

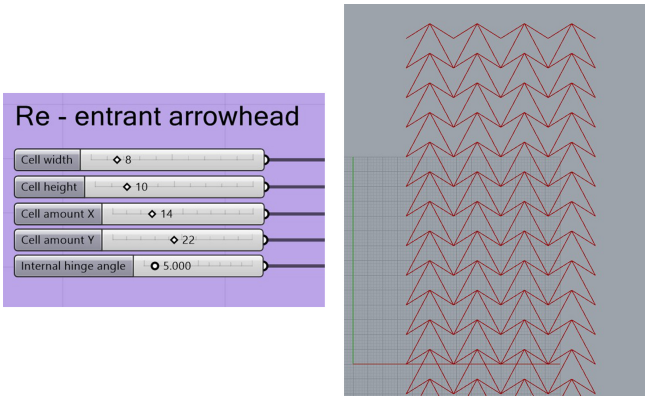


Figure 2: Preview of the auxetic re-entrant arrowhead elastomer toolpath created with a parametric code in Grasshopper for Rhino 3D, including the auxetic cell parameter sliders.

4.2 Heat-bonded fabric lamination

Flexible heat-bonding stabilizers allow for seam stabilization that prevents fabric deformation during sewing, as well as enabling laser-cut finishes that eliminate fabric bulk caused by the need for hems and linings. They are usually either thin continuous strips made of elastic adhesives that are heat-bonded into the fabric or silicone directly printed onto fabric and cured, preventing thread fraying while maintaining fabric flexibility. For the development of this component of the bra, test samples were printed of a solid layer of flexible filament at 0.2, 0.3, and 0.4 mm thick, then heat-bonded to a single layer of the interlock cotton fabric and laser cut at a temperature and speed high enough for cutting without singeing the material. A 0.2 mm thick layer of material was determined sufficient for stabilization and to achieve a laser-cut finish without overly stiffening or thickening the fabric. The heat-bonding stabilizer was modeled in Rhino and printed at a 2-centimeter width on the edges of the pattern pieces, including the edges of the nursing openings of the inner bra cup. To maintain the shape of the printed material during removal from the printing bed to the heat press and laser-cut registration, a frame was added around the shape (Figure 3).

4.3 3D printed clips and adjustment sliders

The third components fabricated were the adjustment sliders for the straps and under bust band, as well as the nursing clips that enable the wearer to remove the outer bra cups for nursing and return them comfortably with one hand. These are components commonly made of stiff synthetic plastic as well as metal for hook-and-eye fasteners, therefore the stiff bioplastic filament Nonoilen was used for FDM printing. While a very small size and thickness of these components is preferred for wearer comfort, the layer-based technology of FDM printers makes it a challenge to create sufficient strength on that scale. Therefore, the nursing clips and adjustable sliders were modeled in Rhino and fabricated for the straps and the under bust at a 3 mm thickness, allowing for sufficient durability Figure 4.

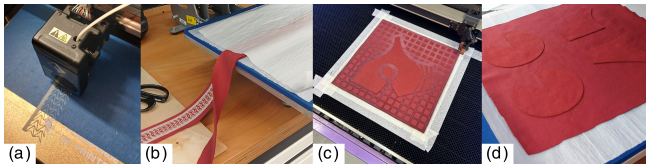


Figure 3: (a) Printed auxetic geometries for the straps and under bust, (b) Heat bonding the auxetic structure between two layers of cotton interlock knitted fabric, (c) Laser cutting the pattern pieces with the heat-bonded stabilizer on the edges and nursing holes, (d) Heat bonding two layers of the bra side panels.

4.4 Case study – Gaia nursing bra

The final prototype of the nursing bra was named "Gaia", in reference to the mythological figure of "Mother Earth", connecting the act of a mother nursing to the regenerative biological cycle of the biodegradable materials used. The bra is adjustable with the sliders on both the under bust and cross-back straps, and the nursing clips with the nursing opening on the inner bra enable single-hand opening and closing for breastfeeding (see Figure 1). The pattern is constructed of an inner bra with nursing holes and an outer bra, all having a laser-cut finish (Figure 4). The side panels of the bra connecting the cups to the under bust band have been heat-bonded together to eliminate excess seams while maintaining a cohesive thickness with the cups (Figure 3).

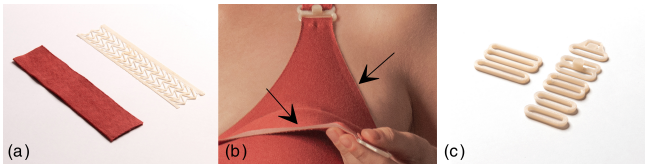


Figure 4: (a) Printed auxetic elastic band with and without the cotton knit fabric encapsulation, (b) Laser cut edge for eliminating hems and reducing bulk for the nursing mechanism, (c) 3D printed slider adjusters and nursing clips.

5 Discussion

The components developed to replace current materials achieved a functional and wearable prototype fitted to a model Figure 1. The development process of the auxetic elastic bands included repeated prototyping to achieve the optimal elasticity and stability for the different bra components, with the most noticeable differences deriving from internal hinge angle, cell scale, and layer amount. The stiff components were limited in terms of their durability due to the FDM printer limitations, but the printed material chosen is also available for injection molding, which could produce smaller and more resilient components for industrial production in a later stage of the design process.

Future work on these design methods will include the development of a fully parametric tool for garment designers, allowing for rapid prototyping of custom elasticated components directly onto a digital pattern. This would allow not only for various iterations of continuous elastic bands, but also precise pattern placement for any shape and variable elasticity. By integrating vector garment pattern pieces, both the auxetic elastomer and the heat-bonding stabilizer can be applied parametrically and translated into G-code for printing.

6 Conclusions

The design process of the nursing bra was driven by its materials as well as the fabrication methods for processing them, in order to achieve a functional and complex garment that is not only biodegradable in industrial compost but is also scalable for high-volume production. The auxetic component can be further developed with different types of knitted textiles and flexible filaments, with results that can be adapted by tuning the cell scale and hinge angle. Creating these tools for designers can help accelerate biodegradable material integration as a part of the garment design process, paving the way to fully compostable garments that don't sacrifice functionality and are fit for industrial production.

References

[1] Wazir Aitizaz Ahsan, Adnan Hussain, Chitsan Lin, and Minh Ky Nguyen. 2023. Biodegradation of Different Types of Bioplastics through Composting—A Recent Trend in Green Recycling. 13, 2 (2023), 294. doi:10.3390/catal13020294

[2] Madhu Balan P, Johnney Mertens A, and M V A Raju Bahubalendruni. 2023. Auxetic mechanical metamaterials and their futuristic developments: A state-of-art review. 34 (2023), 105285. doi:10.1016/j.mtcomm.2022.105285

[3] HuiYing Bao, Yan Hong, Tao Yan, Xiufen Xie, and Xianyi Zeng. 2023. A systematic review of biodegradable materials in the textile and apparel industry. (2023), 1–20. doi:10.1080/00405000.2023.2212848

[4] Sabrina Bianchi, Flavia Bartoli, Cosimo Bruni, Cristina Fernandez-Avila, Laura Rodriguez-Turienzo, Jorge Mellado-Carretero, Daniele Spinelli, and Maria-Beatrice Coltelli. 2023. Opportunities and Limitations in Recycling Fossil Polymers from Textiles. 3, 2 (2023), 120–148. doi:10.3390/macromol3020009

[5] Emanuel Boschmeier, Vasiliki-Maria Archodoulaki, Andreas Schwaighofer, Bernhard Lendl, Wolfgang Ipsmiller, and Andreas Bartl. 2023. New separation process for elastane from polyester/elastane and polyamide/elastane textile waste. 198 (2023), 107215. doi:10.1016/j.resconrec.2023.107215

[6] Christina Bremer, Bran Knowles, and Adrian Friday. 2022. Have We Taken On Too Much?: A Critical Review of the Sustainable HCI Landscape. In *CHI Conference on Human Factors in Computing Systems* (New Orleans LA USA, 2022-04-27). ACM, 1–11. doi:10.1145/3491102.3517609

[7] Jack Forman, Mustafa Doga Dogan, Hamilton Forsythe, and Hiroshi Ishii. 2020. DefeXtiles: 3D Printing Quasi-Woven Fabric via Under-Extrusion. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event USA, 2020-10-20). ACM, 1222–1233. doi:10.1145/3379337.3415876

[8] Maas Goudswaard, Abel Abraham, Bruna Goveia Da Rocha, Kristina Andersen, and Rong-Hao Liang. 2020. FabriClick: Interweaving Pushbuttons into Fabrics Using 3D Printing and Digital Embroidery. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven Netherlands, 2020-07-03). ACM, 379–393. doi:10.1145/3357236.3395569

[9] Daniel Groeger and Jürgen Steimle. 2019. LASEC: Instant Fabrication of Stretchable Circuits Using a Laser Cutter. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow Scotland Uk, 2019-05-02). ACM, 1–14. doi:10.1145/3290605.3300929

[10] Mira Alida Haberfellner, Samuel Zühlke, Leopold Boess, Kathrin Probst, and Michael Haller. 2024. Threads of Traceability: Textile IDs in the Fabric of Sustainable Fashion. In *Designing Interactive Systems Conference* (Copenhagen Denmark, 2024-07). ACM, 3063–3078. doi:10.1145/3643834.3661531

[11] Lon Åke Erni Johannes Hansson, Teresa Cerratto Pargman, and Daniel Sapiens Pargman. 2021. A Decade of Sustainable HCI: Connecting SHCI to the Sustainable Development Goals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama Japan, 2021-05-06). ACM, 1–19. doi:10.1145/3411764.3445069

[12] Elvin Karana, Bahareh Barati, Valentina Rognoli, and Anouk Zeeuw van der Laan. 2015. Material Driven Design (MDD): A Method to Design for Material Experiences. 9, 2 (2015), 35–54. https://research.ebsco.com/linkprocessor/plink?id=7e36bd40-daff-3f2a-980f-a6346789ec89

[13] Mina Konaković, Keenan Crane, Bailin Deng, Sofien Bouaziz, Daniel Piker, and Mark Pauly. 2016. Beyond developable: computational design and fabrication with auxetic materials. 35, 4 (2016), 1–11. doi:10.1145/2897824.2925944

[14] Holly McQuillan and Elvin Karana. 2023. Conformal, Seamless, Sustainable: Multimorphic Textile-forms as a Material-Driven Design Approach for HCI. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg Germany, 2023-04-19). ACM, 1–19. doi:10.1145/3544548.3581156

[15] Robert Pettys-Baker, Heidi Woelfle, and Brad Holschuh. 2023. Robo-Coverstitch: Coverstitched Auxetic Shape Memory Actuators For Soft Wearable Devices. In *Proceedings of the 2023 International Symposium on Wearable Computers* (Cancun, Quintana Roo Mexico, 2023-10-08). ACM, 109–113. doi:10.1145/3594738.3611374

[16] Benjamin Piribauer and Andreas Bartl. 2019. Textile recycling processes, state of the art and current developments: A mini review. 37, 2 (2019), 112–119. doi:10.1177/0734242X18819277

[17] Michael L. Rivera, Melissa Moukperian, Daniel Ashbrook, Jennifer Mankoff, and Scott E. Hudson. 2017. Stretching the Bounds of 3d Printing with Embedded Textiles. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver Colorado USA, 2017-05-02). ACM, 497–508. doi:10.1145/3025453.3025460

[18] Mehrdad Seifali Abbas-Abadi, Brecht Tomme, Bahman Goshayeshi, Oleksii Mynko, Yihan Wang, Sangram Roy, Rohit Kumar, Bhargav Baruah, Karen De Clerck, Steven De Meester, Dagmar R. D'hooge, and Kevin M. Van Geem. 2025. Advancing Textile Waste Recycling: Challenges and Opportunities Across Polymer and Non-Polymer Fiber Types. 17, 5 (2025), 628. doi:10.3390/polym17050628

[19] Haruki Takahashi and Jeeeun Kim. 2019. 3D Printed Fabric: Techniques for Design and 3D Weaving Programmable Textiles. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans LA USA, 2019-10-17). ACM, 43–51. doi:10.1145/3332165.3347896

[20] Daniela Ghanbari Vahid, Lee Jones, Audrey Girouard, and Lois Frankel. 2021. Shape Changing Fabric Samples for Interactive Fashion Design. In *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Salzburg Austria, 2021-02-14). ACM, 1–7. doi:10.1145/3430524.3440633

[21] Eldy S. Lazaro Vasquez and Katia Vega. 2019. Myco-accessories: sustainable wearables with biodegradable materials. In *Proceedings of the 23rd International Symposium on Wearable Computers* (London United Kingdom, 2019-09-09). ACM, 306–311. doi:10.1145/3341163.3346938

[22] Adrian Wagner, Paul Miles Preuschoff, Philipp Wacker, Simon Voelker, and Jan Borchers. 2023. FabricFaces: Combining Textiles and 3D Printing for Maker-Friendly Folding-Based Assembly. In *Extended Abstracts of the 2023 CHI Conference*

on Human Factors in Computing Systems (Hamburg Germany, 2023-04-19). ACM, 1–7. doi:10.1145/3544549.3585854

[23] Tongyan Wang, Hanwen Zhao, Yusuf Shahpurwala, Megan Hofmann, and Jennifer Mankoff. 2025. KnitA11y: Fabricating Accessible Designs with Machine Knitting. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Yokohama Japan, 2025-04-26). ACM, 1–9. doi:10.1145/3706599.3719709

[24] Shanel Wu and Laura Devendorf. 2020. Unfabricate: Designing Smart Textiles for Disassembly. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, 2020-04-21). ACM, 1–14. doi:10.1145/3313831.3376227